

the postulation of ordinary objects, as Unger did, rather than to an inadequate understanding of vagueness. Similarly, the right moral to draw from Zeno's paradoxes is not that motion is impossible but rather that an adequate philosophical theory of motion is needed; the right moral to draw from the paradox of the liar is not that truth and falsity are incoherent categories, but rather that we need an adequate philosophical theory of truth.

* * *

The goal of this chapter is to find the best account, on balance, of the paradoxes of coincidence. So far I have provided arguments against various three-dimensional theories. While I have not considered every such theory, I hope I have considered the most plausible ones. The worm theory, on the other hand, seems to resolve the paradoxes: though the worm theorist must admit coincidence, she can explain its possibility. Of course, the worm theory has its detractors, but their arguments will be answered in chapter 6. The worm theory is therefore the front-runner of the theories considered so far. But I will argue in the next section that there is a four-dimensionalist alternative to the worm theory that does even better, by avoiding coincidence altogether.

8. The Stage View¹⁴⁸

“Coinciding objects are no more mysterious than overlapping roads.” In this way the

¹⁴⁸ See my Sider 1996 for an earlier presentation of the stage view. Related views are discussed briefly in Forbes 1985, pp. 188-189, Perry 1972, and van Inwagen 1990b.

worm theorist answers the *metaphysical* objection that coincidence is impossible. But a semantic objection lingers. Even if we knew that fission was about to occur, we would not *say* that there are two persons before us. It seems wrong to *say* that there are two statue shaped objects before us, the statue and the lump; the more natural thing is to say that there is just one.

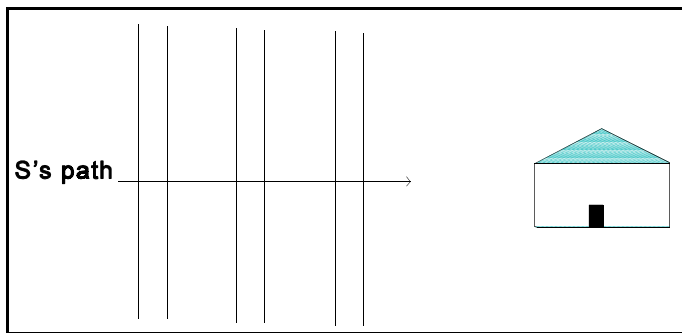
In response, Lewis (1983a, p. 64) defends a revisionary theory of counting. If roads A and B coincide over a stretch that S must cross, when she asks how many roads she must cross to reach her destination it would be appropriate to say “one”. In counting the roads we go through the roads and count off positive integers, as usual. But we do not use a new number for each road — rather, we use a new number only when the road to be counted fails to bear a certain relation to the roads we have already counted. In this case, the relation is that of identity along S’s path, which is born by one road to another iff they both cross S’s path and share sections wherever they do. Say that persons are identical-at-*t* iff their stages at *t* are identical. Counting by the relation identity-at-*t*, there is only one person in the room.

I doubt this procedure of associating numbers with objects is really *counting*. Part of the meaning of ‘counting’ is that counting is by identity; ‘how many objects’ means ‘how many numerically distinct objects’. Counting must be by identity when we count objects not in time (numbers, for example), and surely we count persons in the same sense in which we count numbers. Moreover, the intuition that just one person is present before fission arguably remains even if one *stipulates* that counting is to be identity.

The example of counting one road while giving directions, however, is designed to show that we *do* sometimes count by relations other than identity. I grant that we would indeed say that S needs to cross one road, but I would prefer to say that we have counted *road segments* by

identity. What matters to S is how many road segments she must cross, and we have told her: one. Granted, the question was about roads. But it is plausible that the predicate ‘road’ sometimes applies to road segments rather than “continuant” roads. Whether road segments or continuant roads are meant depends on the speaker’s interests (and may sometimes be indeterminate).

I support this view with an additional example. Suppose S is walking to the farm. As far as we know, her path is as shown in the first diagram. If she asks: “how many roads must I cross



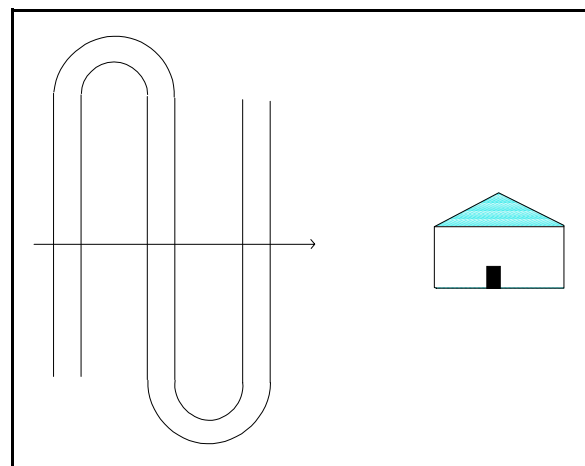
Roads 1

to get to the farm?”, we will answer “three”.

But suppose that, unknown to us, because of their paths miles away the “three” roads are connected, as in the second diagram. In a sense, she only

crosses one road (albeit three times). If we count continuant roads, we count one road that she crosses, whether we count by identity or by

identity-across-S’s-path. But I think we gave the correct answer when we said “three”. We told her what she wanted to know: the number of road segments she needed to cross. If someone later asked me for directions, my short answer would still be “three”. I might add “actually, you cross one road three times”. This might indicate



Roads 2

that her question was ambiguous: does she want to know the number of roads or the number of road segments? But the first answer was satisfactory, for it is likely that she is more interested in road segments.

This way of understanding the case of directions is, I believe, more attractive than counting by relations other than identity. Lewis cites the roads case as evidence that we sometimes count by relations other than identity, but the evidence does not require Lewis's explanation. Counting is by identity.

I suggested that in the case of the winding road, 'road' meant road-segment. The analogous view in the case of persistence would be that the predicate 'person' often — indeed, usually — applies to person stages, not space-time worms. Then, we would never have “over-counts”, since in any possible case, no matter what episodes of fission or fusion the future or past holds, the current count of persons would be determined by the number of person stages present. The view should not be confined to persons. I have been using the term “continuant” for the objects we typically discuss, name, quantify over, and discuss, whatever those objects turn out to be. The view I am suggesting is that all continuants are stages. I call this view the stage view. The opposing four-dimensionalist view is the “worm view”, according to which continuants are temporally extended spacetime worms. So far, the point in the stage view's favor is that it provides the correct “count” in the fission case. It also provides the correct counts in other cases of apparently coinciding objects (Tibbles and Tib, statues and lumps, and so on). All these cases involve distinct spacetime worms that share a single stage in common. Since on the stage view the names for the continuants involved name the shared stage rather than the distinct worms, the stage theorist does away with distinct coinciding continuants.

Given unrestricted mereological composition, I grant the existence of all the worm theorist's worms. My ontology is therefore the same as the worm theorist's: four-dimensionalism. I therefore admit the existence of coinciding entities, for given unrestricted composition, space-time worms that share temporal parts automatically follow. But I deny that these or any other space-time worms are *continuants*; they are not ordinarily named or quantified over. Coincidence between space-time worms is *metaphysically* unobjectionable, as was argued in §2 above. The remaining semantic objection was that coincidence does not seem to be the correct description of the puzzle cases. But semantic intuitions pertain to everyday predicates, (such as 'person' and 'statue'), under which spacetime worms do not fall. Moreover, these worms are (typically) not in the range of our quantifiers. Thus, the stage theorist can deny that any of the objects over which we typically quantify ever coincide.

Peter Unger's "problem of the many"¹⁴⁹ might be thought to undermine my claim that the stage view accounts for counting better than the worm view. In the fission case, for example, I say that at any given time before fission there is just one person stage in the vicinity. But a person is surrounded by various particles of matter that are neither definitely part of, nor definitely not part of, that person. Hence, pre-fission, there are many fusions of particle stages, mostly overlapping but differing slightly from each other around the periphery of the person in question, each with approximately the same properties relevant to personhood. How then can I say that there is just one person stage there?

Fortunately, the problems about counting raised by the problem of the many are orthogonal to those raised by the paradoxes of coincidence. Once the problem of the many is

¹⁴⁹ See Unger 1980, Lewis 1993 and Hudson (forthcoming) chapters 1 and 2.

solved in some acceptable way, the coincidence-based problems remain, and can still motivate the stage view. Consider for example, the two solutions advanced by Lewis (1993), as applied to the case of a person alone in a room. According to the first approach, that of supervaluations, ‘there is just one person in the room’ is true since on any admissible precisification of the vague term ‘person’, exactly one of the candidate persons falls under its extension. According to the second, identity is the limiting case of mereological overlap; each of the many candidates is indeed a person; though these persons are numerically distinct from each other they extensively overlap, and so each is *almost* identical to the rest. Though it is not strictly true that there is just one person, it is almost true that there is just one person, and that is good enough. Each of these solutions seems promising. When applied to the case of a person stage, each generates the result that, at any time before fission, (it is at least almost true that) there is just one person stage. Thus, if persons are person stages, we have one person. But if persons are spacetime worms there are two persons, given either of Lewis’s solutions to the problem of the many. On any precisification of ‘person’, there are two worms that count as persons. (If some precisifications counted only one of the two overlapping worms as a person then a *post*-fission utterance of “there are two persons in the room” would not turn out true — an unacceptable result.) Moreover, while the two worms overlap, the overlap is nowhere near total; hence the worms are not almost identical.

The stage view thus has the virtue of generating the proper counts in the cases of coincidence. But is it an acceptable view? The first and most obvious objection is that it denies persistence over time. How could continuants be stages; stages don’t *continue*? If persons are instantaneous stages, then no person lasts more than an instant. However, as I will develop it, the

stage view includes a counterpart theory of *de re* temporal predication, according to which instantaneous stages may nevertheless have temporal properties such as being F in ten minutes.

According to my temporal counterpart theory, the truth condition of an utterance of ‘Ted was once a boy’ is this: there exists some person stage *x* prior to the time of utterance, such that *x* is a boy, and *x* bears the temporal counterpart relation to Ted.¹⁵⁰ Since there is such a stage, the claim is true. Despite being a stage, Ted *was* a boy; he has the historical property of once being a boy. Similar accounts of ‘I was a boy 20 years ago’ and ‘I lived in Massachusetts in 1990’ may be given, as having the truth conditions, respectively, that I have a temporal counterpart 20 years prior to the time of utterance that is a boy, and that I have a temporal counterpart in 1990 that lives in the 1990 counterpart of Massachusetts.¹⁵¹

The temporal counterpart relation is the same relation used by the worm theorist to unite the stages of space-time worms. Also known as the “genidentity relation”, the “unity relation”,

¹⁵⁰ In section VII of Sider 1996 certain semantic complications are explored in more detail. A proper name denotes different objects relative to different times. There then arises a distinction between a *de re* and a *de dicto* reading for temporal predications. “I was *F*” can be taken to be *de re* about my present stage; it is to these uses that the counterpart theoretic account is directed. But “Socrates was *F*” cannot be, since Socrates has no present stage; the truth condition here is that at some time in the past, the referent of ‘Socrates’ with respect to that time — a stage — is *F*. Mixed cases are also possible. In fact, some of the sentences I discuss later in this chapter should be taken in this mixed way, for example the sentence “Ted will be identical to Fred”. This should be interpreted as *de re* with respect to ‘Ted’, but *de dicto* with respect to ‘Fred’; that is, it should be taken as having the following truth conditions: the current referent of ‘Ted’ has a future counterpart that is identical to the referent of ‘Fred’ with respect to that time.

¹⁵¹ Note that the metalanguage locutions ‘is a boy’ and ‘lives in’ in these truth conditions are intended to express tenseless predication, not present-tense predication. The stage theory is an account of the truth conditions for a tensed language like English, but it is given in a tenseless language (though sometimes I will slide into a tensed metalanguage for ease of expression.)

the “I-relation”, and so on,¹⁵² it may be analyzed in some way (in the case of persons perhaps in terms of memory or bodily continuity), or taken as primitive; the stage theorist has no particular commitment to any of these alternatives.

There is a close analogy here with modal counterpart theory. According to modal counterpart theory, an object, *x*, has the property possibly being F iff some counterpart of *x* in some possible world has F. The temporal operator ‘was’, and also other temporal operators like ‘will be’, ‘will be at *t*’, etc., are analogous to the modal operator ‘possibly’. (The analogy is only partial, for there are no modal analogs of metrical tense operators like ‘will be in 10 seconds’.)

This analogy will be important in what follows. For example, it indicates the line of defense against the following objection: “According to the stage view, statements that look like they are about what once happened to me are *really* about what once happened to someone else. And that’s absurd.” John Perry gives something like this objection to a view something like my temporal counterpart theory:¹⁵³

...[on (a view like) the stage view] the little boy stealing apples is *strictly speaking* not identical with the general before me.... [The stage view] denies what is clearly true: that when I say of someone that he will do such and such, I mean that he will do it. The events in my future are events that will happen to me, and not merely events that will happen to someone else of the same name.

Compare Saul Kripke’s (1972, p. 45) famous objection to Lewis’s modal counterpart

¹⁵² See the introduction to Perry 1975 (‘unity relation’) and Lewis 1983a (‘I-relation’). Carnap (1967, §128 and §159) uses ‘genidentity’, attributing the term to Kurt Lewin (1923).

¹⁵³ Perry 1972, pp. 479, 480. Nathan Salmon also appears to be giving an objection of this sort to a theory like the stage view in Salmon 1986 pp. 97-99.

theory:

[According to modal counterpart theory,]...if we say ‘Humphrey might have won the election (if only he had done such-and-such)’, we are not talking about something that might have happened to *Humphrey* but to someone else, a “counterpart”. Probably, however, Humphrey could not care less whether someone *else*, no matter how much resembling him, would have been victorious in another possible world.

But according to Lewis, counterpart theory does *not* imply that we are talking about what might have happened to someone other than Humphrey. Granted,

Counterpart theory does say ... that someone else - the victorious counterpart - enters into the story of ... how it is that Humphrey might have won.

But what is important is that Humphrey himself has the modal property:¹⁵⁴

Thanks to the victorious counterpart, Humphrey himself has the requisite modal property: we can truly say that *he* might have won.

The reply is the same in the temporal case. Perry is wrong to say that the stage view denies that ‘You will do it’ means that *you* will do it. ‘Ted was once a boy’ attributes a certain temporal property, the property of once being a boy, to me, not to anyone else. Of course, the stage view does analyze my having this property as involving the boyhood of another object, but *I* am the one with the temporal property, which is the important thing. The stage view is consistent with

¹⁵⁴ See Lewis 1986a, p. 196. See also Hazen 1979.

stages having temporal properties; it's just that temporal properties are given a counterpart-theoretic analysis.

It is a common complaint that modal counterpart theory does not *really* allow individuals to have modal properties.¹⁵⁵ Perhaps a residue of dissatisfaction remains in the temporal case, based on something like this sentiment. But I reject the complaint, both against modal and temporal counterpart theory. What is certain is that modal and temporal properties, however analyzed, apply to everyday objects. The *theory* about the proper analysis of such properties is another matter. If a theory can save the appearances — if it can account for the distribution of temporal and modal properties we take things to have; if it does not require an objectionable ontology or ideology; and if it meets these desiderata better than competing theories — then it is reasonable to adopt.¹⁵⁶ I do not say that intuitions about theoretical analyses carry no weight at all, only that they are not sacrosanct. Indeed, I partially based my rejection of the worm view on theoretical intuitions about counting. I grant that my analysis of tensed predication is unexpected, but this is not a *decisive* consideration, and is outweighed by the stage view's benefits.

The matter may profitably be viewed in light of the best candidate theory of content, as presented in the introduction to this book. Ordinary beliefs about persistence and modality lay down a persistence-role and a modal-role. Candidates for being meant by talk of persistence and modality satisfy these roles to greater or lesser degrees. I say that the counterpart theoretic candidates satisfy the roles best. (In the temporal case this is in large part because of the

¹⁵⁵ See, for example, Plantinga 1974, chapter 6, and Salmon 1981, p. 235, note 16.

¹⁵⁶ See also Hazen 1979 pp. 320-324 and Forbes 1987, p. 143.

attractive stage-theoretic solution to the paradoxes of co-location to which I will return shortly; this would need to be argued in the modal case.) Moreover, the counterpart-theoretic candidates are just as intrinsically eligible to be meant as their competitors. (In the temporal case, assuming four-dimensionalism is true the competing candidate is worm-theoretic persistence, and it is clear that neither of these candidates carves nature at its joints more than the other. In the modal case this claim about eligibility would need to be argued.) The counterpart-theoretic candidates are thus the best candidates to be meant by talk of persistence and modality. And that is all it takes to be meant.

The difference between temporal counterpart theory, on one hand, and mereological essentialism and nihilism, on the other, may now be brought more clearly into focus. Recall the trumping argument against these views. *Entia successiva*-persistence would be a weaker candidate for being meant by persistence-talk than the *genuine* persistence of mereologically constant objects. For the nihilist, persistence of simples would be genuine persistence; statements about persisting macroscopic tables and chairs would turn out false. But assuming four-dimensionalism is true, counterpart-theoretic persistence is as good as it gets, and is thereby the best candidate, and is thereby true persistence. It might be argued that worm-theoretic persistence would trump stage-theoretic persistence, but it is precisely the arguments of the present section that show this is not the case. Stage-theoretic persistence best accounts for our intuitions about counting and other intuitions to be discussed below. At any rate, the stage theorist need admit no distinguished class of continuants whose persistence is more “ideal” than that of others.

The stage theorist can therefore account for the possession of tensed properties by stages.

I must concede, however, that tenseless statements of “cross-time” identity are false, for example ‘I am identical to a young child’. This strikes me as no bad thing, given that the tensed sentence ‘I am identical to something that *was* a young child’ *does* turn out true. More simply: I was a young child. That is the way we say it in English, and that is what is important to vindicate.

I therefore find the Kripke/Perry objection to temporal counterpart theory unpersuasive. There is, however, a different objection that I take very seriously, which ultimately seems to require a concession. Certain sentences involving “timeless counting” are ill-handled by temporal counterpart theory, for example “fewer than two billion persons have set foot in North America throughout history”. The problem, of course, is that if ‘person’ refers to person stages this sentence will turn out false, since more than two billion (indeed, infinitely many if time is dense) person stages have set foot in North America throughout history.

As I say, a concession seems required. Some — perhaps most — talk of continuants receives a stage-theoretic analysis, but in some cases we need a worm-theoretic account after all. My considered view, therefore, is that a sort of indeterminacy or ambiguity should be postulated. Sentences involving counting of the non-timeless variety, for example “there is one person in the room before me now”, receive a stage-theoretic analysis, as do certain sentences to be discussed below, since this analysis makes the best sense of our intuitions about those sentences. But in cases like that of timeless counting, a worm-theoretic analysis seems required. The concession perhaps makes the stage view a little less attractive since, arguably, candidate semantics that postulate this sort of ambiguity or indeterminacy seem, other things being equal, weaker than

candidates that do not.¹⁵⁷ Nevertheless, the stage view's advantages outweigh this defect.

A further objection to the identification of continuants with stages is that stages do not exist long enough to have many of the properties we attribute to continuants. The sentence 'Ted believes that four-dimensionalism is true' attributes, according to the stage theorist, a belief to my current stage. But how can an instantaneous stage believe anything? Beliefs take time.

Having a belief does indeed require *having had* certain features in the past. This is not inconsistent with the stage view, which interprets the past having of the relevant features as amounting to having temporal counterparts that have those features. In order to have a belief, a stage must stand in an appropriate network of counterpart relations to other stages with appropriate features. Thus, the property having a belief is a highly relational property. It nevertheless can be instantiated by instantaneous stages. Notice that I do not claim to be able to *analyze* relational properties of this sort in terms of intrinsic features of stages and the counterpart relation. Such questions about analysis are no more my responsibility than anyone else's.

A final question one might have about the stage view is whether it can be reconciled with the special theory of relativity. What counts as a *temporal* stage of a space-time worm varies with frame of reference. Clearly the stage theorist must in some way relativize his central claim, that continuants are temporal stages, to frames of reference.

Suppose a speaker refers to an object. Is the object to be identified with a temporal stage

¹⁵⁷ Jeremy Pierce pointed out in conversation that an ambiguity theory may have more attractive consequences than either the worm or stage view in the case of a time traveler talking to his former self. How many persons are involved in the conversation? The stage theorist says *two*, the worm theorist *one*. But perhaps our intuitions are indeterminate between one and two, just as the ambiguity theory predicts they should be.

construed relative to the *object's* or the *speaker's* frame of reference? Not the former, I think.

For suppose a speaker makes a tensed utterance about two objects at once, for example:¹⁵⁸

Next year it WILL be the case that Greg Maddux strikes out Mike Piazza.

'Greg Maddux' and 'Mike Piazza' refer to objects M and P that count as temporal slices, relative to a pair of frames of reference F_M and F_P , of the Maddux and Piazza worms at the time of the sentence's utterance. In addition, a frame of reference must be chosen to interpret the tense operator 'WILL'. The counterpart-theoretic truth condition for this sentence, as uttered at t , is that M and P have temporal counterparts at some time *future* to t , the first of which strikes out the second. But futurity (and the notion of "time t ") must be relative to some frame of reference, F_{WILL} . So there are three frames of reference floating about: F_M , F_P , and F_{WILL} . It would be very odd to choose either of the object-stages by a different frame of reference than that used to evaluate the WILL operator. Thus it should turn out that $F_M = F_{WILL}$ and $F_P = F_{WILL}$, and hence that $F_M = F_P$. Thus, the object stages M and P cannot be construed as temporal parts relative to the frames of reference of Maddux and Piazza, since Maddux and Piazza may well be moving in different frames of reference. The easiest way to ensure that $F_M = F_P = F_{WILL}$ is to set each identical to the frame of reference of the speaker.

The central claim of the stage view is supposed to be that *a speaker refers to stages of worms sliced at the time of utterance*. If we could assume that every speaker follows a one-dimensional worldline through spacetime, and that each of her utterances occur at a single point

¹⁵⁸ Maddux and Piazza are professional baseball players.

along this worldline, then the stage view's central claim could be given a clear sense: when a speaker makes an utterance at a point p of spacetime, she refers to temporal stages of worms at the time that contains p , where 'time' and 'temporal stage' are relativized to her frame of reference when she passes through p . But each of the assumptions is an idealization. Utterances cannot in general be associated with single points in spacetime since they take time and cannot be spatially confined to a point. Moreover, speakers themselves follow four-, not one-, dimensional paths through spacetime. So just how is the central claim of the stage view to be understood?

The solution is to abstract away from utterances. A stage theorist should provide an account of a somewhat theoretical notion, that of a sentence type's being true as uttered at a time t , understood relative to frame of reference F . The stage theorist should claim, for example, that the sentence type 'Ted will be bald', as uttered at t , interpreted relative to F , is true iff the Ted-stage at t , relative to F , has a temporal counterpart in the future, relative to F , that is bald. How, then, is this theoretical notion to be *applied*? How can the stage theorist leave the ivory tower and make claims about truth values for real live utterances of sentences? The stage view can be applied provided we are willing to live with a lot of harmless indeterminacy. The principle of application is as follows. For any utterance u of a sentence type T , there will be a range of spacetime points where u could be regarded as occurring, and a range of frames of reference that could be regarded as the speaker's on the occasion of making utterance u . Call any pair $\langle F, p \rangle$ of one of these frames of references, F , and points, p , an "admissible pair". For any admissible pair $\langle F, p \rangle$, p counts as being at a certain time, t , relative to F . The stage view's account of the theoretical notion then supplies a truth value for T as uttered at t relative to F . Thus, for any admissible pair, the stage theory generates a truth value for u . For most utterances, the truth

value will be constant for all admissible pairs, and in those cases we can harmlessly approximate the truth by saying that the utterance has this truth value *simpliciter*. When the truth value does not remain constant for all admissible pairs, it seems right to say that the utterance has no determinate truth value.

So much for answering objections. Let us now systematically examine the stage-theoretic account of the paradoxes of coincidence, beginning with the statue and the lump of clay. It is a great advantage of the stage view that it vindicates the natural reaction to such cases: the statue should not be distinguished from the lump today just because “they” will differ tomorrow. The stage theory thus preserves the core insight of the thesis of temporary identity (§5) while avoiding its pitfalls. Identifying lumps and statues with stages implies the desired claim that the statue is identical to the lump, at the time of coincidence.¹⁵⁹ But we also want to say that “they” differ in their histories:

- (L) The lump will exist tomorrow
- (S) The statue will not exist tomorrow

How can (L) and (S) be true, if the statue *is* the lump?

¹⁵⁹ Insuring that ‘the statue = the lump’ turns out true requires some subtlety if we solve the problem of the many using supervaluations. If the supertruth condition for this sentence begin simply with: “for every admissible precisification of ‘statue’ and every admissible precisification of ‘lump’, ...”, the sentence will not be supertrue since some precisifications of ‘lump’ contain in their extensions a different one of the many stages in the region than do some precisifications of ‘statue’. The supertruth conditions should reflect a kind of “penumbral connection” (in the sense of Fine 1975) between ‘the statue’ and ‘the lump’: “for every admissible precisification of ‘statue’ and every *corresponding* admissible precisification of ‘lump’ ...”.

The answer is that the expressions attributing temporal properties are ambiguous, so there is no one property that (L) predicates of the lump but (S) withholds from the statue. ‘Existing tomorrow’ expresses different properties in different contexts since the relevant temporal counterpart relation can vary from context to context.

A temporal counterpart relation specifies what sort of “continuity” a thing must exhibit over time in order to continue to exist. But there are different kinds of continuity on which speakers focus. A certain kind of continuity is destroyed when a statue gets squashed, since the item has not retained a statue shape. Let us say that the *statue counterpart relation* does not hold between the statue stage before and the lump stage afterwards. But there is another kind of continuity that is not destroyed: the holding of a *lump-of-matter counterpart relation* underwrites our intuitive judgment that the same lump is present.

The lump counterpart relation is used in a typical interpretation of (L) because we denote the object in question using the term ‘the lump’. (L) thus means that the lump (= the statue) has a lump counterpart tomorrow. When we denote that same object by the term ‘the statue’, this triggers the use of the statue counterpart relation. (S) means that the statue (= the lump) has no statue counterpart tomorrow. This all is consistent with the identity of the statue and the lump. Here I have again exploited the analogy between the stage view and modal counterpart theory: Lewis (1971) has accounted for contingent identity in a parallel way using multiple counterpart relations.¹⁶⁰

¹⁶⁰ I claim that I am identical to an *instantaneous* stage, and also that I will exist for more than instant — how can I have it both ways? When I say that a “stage” is instantaneous and so will not exist tomorrow, I am denying that it is *stage-counterpart* related to any stage in the future; the stage-counterpart relation is that of identity. When I say that “I” will exist tomorrow, I mean that I have *person-counterparts* that exist tomorrow.

The case of Tibbles and Tib is similar. The identification of Tibbles with Tib after detachment of the tail seems precluded by the apparent truth of the following claims:

Tibbles once had a tail
Tib never had a tail

The stage-theoretic solution is that the first sentence claims that Tibbles has *cat* counterparts with tails, whereas the second claims that Tib has no *torso*-counterparts with tails.

The case of fission has a particularly nice resolution. The original puzzle was that both “branches”, Fred and Ed, seem to matter to the original person, Ted; the principle that identity is what matters then seems to imply that each will be identical to Ted. A resolution appealing to coincident entities was provided, but the stage theorist can do better. He can say that there is only one person before fission, namely Ted; that Ted will be identical to Ed; *and* that Ted will be identical to Fred! This seems like the right thing to say about the case: the person about to divide *will be* the one branch, and it is also true that he will be the other. This does not imply the paradoxical conclusion that the branch persons Fred and Ed are identical to each other. To say that Ted will be identical to Ed is to say that Ted has a future (person-) counterpart that is identical to Ed. More simply, it is to say that Ed is Ted’s counterpart. To say that Ted will be identical to Fred is to say that Fred is Ted’s counterpart. But this does not imply that Ed is Fred, provided a person can have two distinct temporal counterparts at a single time in the future.

This resolution is particularly welcome given a difficulty in Lewis’s resolution of the case, which was brought out originally by Parfit 1976, and which I reconstruct somewhat

differently as follows. The §1 argument for coincidence based on fission (and the arguments from fusion and longevity as well) appealed to (=WM):

(=WM) Person x matters to person y iff x is identical to y

I said that since each branch preserves what matters to *someone*, (=WM) implies that there were two persons all along. But in fact, in addition to the underlined sentence, the following may be argued to be true:

(*) For any person beforehand, x , each of the post-fission persons, Fred and Ed, matters to x

Between *any* initial pre-fission person and either Fred or Ed there hold the same intrinsic relations as in a non-branching case where only a single hemisphere of the original person survives. What matters is preserved in the non-branching case; since mattering is an intrinsic relation it must hold between any pre-fission person in the branching case and either Fred or Ed as well. (*) therefore is true. But (*) is inconsistent with Lewis's account of the fission case, given (=WM). According to Lewis, Ed exists beforehand; by (*), Fred afterwards matters to Ed; by (=WM) we have the conclusion that Fred = Ed, which Lewis must reject since Fred and Ed are, on his view, distinct space-time worms.

In "Survival and Identity", Lewis claims to have reconciled the principle that identity is what matters with the idea that both branches preserve what matters by claiming that the I-

relation (that is, the genidentity, or unity relation for persons) holds between pre-fission stages and stages on *both* branches.¹⁶¹ If the mattering relation in question were a relation over *stages* this would perhaps be adequate, for in that case the thesis that identity is what matters and the principle that each branch preserves what matters would become, not (=WM) and (*), but rather the following theses:

- Stage *x* matters to stage *y* iff stages *x* and *y* stand in the I-relation
- Stages of Fred matter to stages of Ted; and stages of Ed matter to stages of Ted

Given that the I-relation need not be both transitive and symmetric, no contradiction would result. But Lewis's goal was to reconcile the *commonsense* platitude, as he puts it, that identity is what matters, with the idea that each of the branches preserves what matters. If the platitude is to be one of commonsense, it must concern a mattering relation that applies to *persons*. For Lewis, persons are space-time worms, not stages. In that case, the platitude and the claim that both branches matter must be cashed out as (=WM) and (*) above, and the solution collapses.

Lewis's response to Parfit in the postscript to his paper "Survival and Identity", as I understand it, is that (=WM) is false. As he puts it (p. 74):

[A single person stage shared by Ed and Fred before division] does the thinking for both of the continuants to which it belongs. Any thought it has must be shared. It cannot desire one thing on behalf of [Ed] and another thing on behalf of [Fred].

¹⁶¹ Lewis actually discusses the view that relation R is what matters, where R is a relation of psychological continuity and connectedness; but the issues are parallel for any view on which each branch preserves what matters.

I responded to this objection on behalf of Parfit in my 1996; I would like now to take up the matter again. Lewis's claim seems to be that since Ed cannot desire anything uniquely on behalf of himself, we cannot say that Ed, but not Fred, matters to Ed; therefore, (=WM) is false. But this is conceding defeat. The goal was to say that identity is what matters. But this requires that what happens to *another* person cannot matter to me. So if, as Lewis seems to be claiming, *both* what happens after fission to Ed and what happens then to Fred matter to Ed before fission (and also to Fred), then the goal of preserving the commonsense platitude that identity is what matters has not been reached. Lewis's argument seems to be that if we consider his description of the situation we will see that (=WM) is false. But isn't that just to concede that his description of the situation violates the commonsense platitude?

Moreover, we can restate Parfit's argument so that it is invulnerable to Lewis's observations about Ed's desires. My earlier attempt (1996, p. 436) was to argue that we have a concept of rational self-interested concern that is independent of what we are capable of desiring, and therefore that Ed's inability to *desire* anything uniquely on his own behalf is no impediment to Ed's being uniquely concerned, in the relevant sense, with Ed. But such a notion of concern, so divorced from desiring, may be regarded as too distant from ordinary belief to provide a basis for theory choice, so I will introduce a new example. Let us consider fusion, not fission, and let the relevant principles concern responsibility for a past crime, rather than concern. The commonsense platitude that no person can be punished for another's crime must be reconciled with the evident truth that the person who results from fusion seems guilty for crimes committed by either of the branches. And now the relevant attitude — responsibility — is not one that must

be underwritten by anything like desire. There is, therefore, no room for a response like that in Lewis's postscript. Lewis can still, of course, say that once his description of fusion is appreciated, we will see that there is no impediment to punishing the resulting person for crimes of one of the original branches, despite the fact that this means punishing an innocent person (the innocent branch, who now shares stages with the guilty branch; of course, the guilty person is punished along with the innocent one.). But this is simply giving up on, not preserving, the commonsense platitude that a person cannot be justly punished for crimes she did not commit. It would be better to preserve the platitudes.

As argued above, the stage theorist can indeed preserve the platitudes. She must simply hold in the fission case that a stage can have two temporal counterparts at a single future time, and in the fusion case that two distinct stages can share a common counterpart in the future. Stage-theoretic versions of the principles that identity is what matters and that both branches preserve what matters are the following:

- Person *x* matters to an earlier person, *y*, iff *x* was identical to *y*
- Ed matters to Ted; Fred matters to Ted

In the fission case these claims imply that Ed was identical to Ted and that Fred was identical to Ted; it does not follow that Fred is identical to Ed.

The stage theorist can avoid admitting coincident persons in the longevity case by admitting intransitive counterpart relations. Return to Parfit's everlasting bodies, and consider persons P_{1700} , P_{1800} , and P_{1900} , which are, according to the stage theorist, three distinct stages,

existing at times 1700, 1800, and 1900. The puzzle was driven by the assumption that mattering extends over intervals of up to, but not more than, 137 years. The following claims follow:

P_{1800} matters to P_{1700}
 P_{1900} matters to P_{1800}
 P_{1900} does not matter to P_{1700}

Then by the stage-theoretic version of the principle that identity is what matters, it follows that:

P_{1700} will be identical to P_{1800}
 P_{1800} will be identical to P_{1900}
 P_{1700} will not be identical to P_{1900}

But the latter three claims do not violate the transitivity of identity. They are, syntactically, the results of applying tense operators to identity statements; on analysis they become:

P_{1800} is a counterpart of P_{1700}
 P_{1900} is a counterpart of P_{1800}
 P_{1900} is not a counterpart of P_{1700}

The final triad is consistent, provided the counterpart relation can be intransitive.

Finally let us consider the cases of vague and conventional identity. We were driven to accept coincident entities to account for *multiple candidates*, candidates among which singular terms were indeterminate in reference, in the case of vagueness, and equally acceptable but unchosen candidates for reference, in the case of conventionality. The stage view allows us to

provide the requisite candidates but still avoid coincidence: the candidates are different counterpart relations. We can attribute the vagueness in whether the original restaurant B_0 is identical to B_1 or B_2 to indeterminacy in which counterpart relation to use in evaluating claims like: “Restaurant B_1 was restaurant B_0 ”. And we can attribute the element of conventionality in identifying one club with another to the existence of equally good alternatives to the one chosen counterpart relation used in evaluating tensed claims about clubs.

The stage theorist can resist the arguments for coincident entities in all of the puzzle cases we have discussed. All competing theories, as argued above, fail to do away with coincidence in at least some cases, or deny deeply held beliefs about persistence. This, then, is a powerful argument in favor of the stage view, and hence of its presupposed ontology, four-dimensionalism.

A common complaint against the worm theory is that it provides a solution only to the temporal versions of the paradoxes of coincidence, whereas many of the puzzles have interesting modal versions. The worm theory on its own cannot explain the apparent modal differences between Gibbard’s Lump1 and Goliath, since Lump1 and Goliath stage-share at every moment at which either exists and hence seem to be the very same space-time worm. Either the argument in favor of coincidence must be resisted, or some alternate explanation of coincidence must be given. But admitting either of these alternatives would undermine the case for temporal parts, which is in part based on the rejection of alternative responses to the paradoxes of coincidence.¹⁶²

Given the stage view, however, a single unified treatment of the paradoxes, both temporal and modal, is possible. That account is thoroughly counterpart-theoretic. We have already met

¹⁶² See Burke 1994a, pp. 593-594; Rea 1995.

the counterpart-theoretic solution to the modal statue and lump puzzle (chapter 4, §8); the counterpart-theoretic solution to the modal paradox of undetached parts will be discussed in chapter 6, §4. The other puzzles also have familiar modal analogs. All these modal puzzles can be resolved with modal counterpart theory just as the temporal puzzles have been resolved with temporal counterpart theory. Unlike a worm theorist, then, a stage theorist need not “shift gears” when moving to the modal case. It is satisfying that the same medicine cures what appears to be the same disease. Some, of course, do not like counterpart theory, especially modal counterpart theory. Van Inwagen (1990b), as we will see in chapter 6, §4, opposes four-dimensionalism by showing that it implies (modal) counterpart theory! Many share this attitude towards modal counterpart theory; it is a philosophical bogey-man of the first magnitude, as bad as solipsism or logical positivism. But what is the problem? Counterpart theory does not require the modal realism defended by its most visible advocate, David Lewis (1986a); it is consistent with a number of views about the metaphysics of possible worlds and individuals (for example that defended in Sider (unpublished B)).¹⁶³ Set aside this guilt by association; and set aside the Humphrey objection that has already been addressed. What is the big deal?

Perhaps the big deal is, paradoxically, counterpart theory’s greatest strength. Dean Zimmerman has suggested to me that counterpart theory’s flexibility in solving modal and temporal puzzles — the very feature that makes it attractive in the present context — is what lies behind pervasive queasiness about counterpart theory. “So you have conflicting intuitions about *de re* modality or temporality? The conflict is only apparent, since the intuitions concern

¹⁶³ This point is made in Stalnaker 1986 (van Inwagen is fully aware of the point; see his 1990b, p. 254.)

different counterpart relations. In fact, I can produce a counterpart relation corresponding to any intuition you might have!” Perhaps these words of the counterpart theorist make life too easy. Perhaps one of our deeply held beliefs about modality and temporality is that there is always a *single, univocal, non-conventional* answer to the questions “What will happen to this thing tomorrow?” “What might have happened to this thing?” The counterpart theorist must admit that pretty much any answer to these questions could, in principle, be correct, given an appropriate choice of a counterpart relation. Actual spoken languages may well have significant restrictions on what counterpart relations are chosen, and it need not be that an individual speaker may choose whatever counterpart relation she likes. Nevertheless, it seems that a counterpart theorist cannot accept the existence of “deep”, “non-conventional” facts about *de re* persistence and modality.

But the non-existence of such facts is precisely the moral of the puzzles of persistence and their modal analogs. Must these questions have “deep”, univocal, non-conventional answers?:

Will the clay statue survive being squashed?
 After Tibbles’s tail is severed, can the resulting cat say truly that it once had a tail?
 What happens to me after I divide?
 Did Bookbinder’s survive as Bookbinder’s Seafood House or The old original
 Bookbinder’s?

Or consider Terence Parsons’s (1987) pile of trash lying on the road for a day, undergoing partial but not total replacement of constituent refuse. Is the pile of trash at the end of the day the same as the pile of trash there at the beginning? When we consider these cases, our overwhelming

feeling is surely that the answers to the questions depend on how we conceptualize the objects involved, that the world does not force single answers upon us. Any pre-reflective belief to the contrary is a casualty of philosophical reflection, disappearing once the puzzle cases are taken seriously.

If it is counterpart theory that really does the work in solving the paradoxes of coincidence, might four-dimensionalism be dispensed with? Might counterpart theory be combined with three-dimensionalism? We have already met one theory that pretty much amounts to a three-dimensionalist counterpart theory: Chisholm's theory of *entia successiva*. Its problem, as noted above, is that the *real* persistence enjoyed by mereological sums would trump the claim of counterpart-theoretic persistence to be referred to by ordinary talk of persistence.

A certain kind of presentist, on the other hand, could be a temporal counterpart theorist and avoid the trumping argument. Suppose the presentist accepts only primitive "*de dicto*" or "qualitative" tenses — tense operators whose scope cannot contain proper names, demonstratives, indexicals, or quantified-in variables. Ersatz past and future individuals could then be constructed using purely qualitative formulas and the tenses, just as an actualist can construct ersatz possible individuals using qualitative formulas and the modal operators.¹⁶⁴ The open formula '*x* is a farmer & *x* lives on the moon & ...' would represent a future possible individual iff the following tensed claim is true: 'WILL $\exists x(x$ is a farmer & *x* lives on the moon & ...)'. The temporal counterpart relation could then be defined over these ersatz past and future individuals, and sentences ascribing *de re* temporal predication, for example ' $\exists x$ WAS (*x* is a boy)', could be analyzed using temporal counterpart theory.

¹⁶⁴ See Lewis 1986a, pp. 148-150.

Why the restriction to qualitative tenses? Because otherwise the primitive tenses on their own would suffice for the truth value of *de re* temporal predication, and counterpart theory would be superfluous. The primitive tenses would trump counterpart theory's claim to give the meaning of temporal predication, just as did the genuine persistence of 3D sums according to the mereological essentialist. (A similar trumping argument, it seems to me, shows that the primitive modal operators of an actualist counterpart theorist must be *de dicto*.) This sort of presentist would avoid the trumping argument against mereological essentialism, because given the abandonment of primitive *de re* temporal predication, counterpart-theoretic persistence would be "as good as it gets". No presentist I know of accepts this limitation of the tenses, but I see no obstacle to its imposition. Nor do I see any obstacle to this version of counterpart theory other than the underlying presentism. But that, as argued in chapter 2, is obstacle enough.